Improving Mood Classification in Music Digital Libraries by Combining Lyrics and Audio

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ABSTRACT
Mood is an emerging metadata type and access point in music digital libraries (MDL) and online music repositories. In this study, we present a comprehensive investigation of the usefulness of lyrics in music mood classification by evaluating and comparing a wide range of lyric text features including linguistic and text stylistic features. We then combine the best lyric features with features extracted from music audio using two fusion methods. The results show that combining lyrics and audio significantly outperformed systems using audio-only features. In addition, the examination of learning curves shows that the hybrid lyric + audio system needed fewer training samples to achieve the same or better classification accuracies than systems using lyrics or audio singularly. These experiments were conducted on a unique large-scale dataset of 5,296 songs (with both audio and lyrics for each) representing 18 mood categories derived from social tags. The findings push forward the state-of-the-art on lyric sentiment analysis and automatic music mood classification and will help make mood a practical access point in music digital libraries.

Categories and Subject Descriptors
H.3.1 [**Information Storage and Retrieval**]: Content Analysis and Indexing – *indexing methods, linguistic processing.* H.3.7 [**Information Storage and Retrieval**]: Digital Libraries – *systems issues.* J.5 [**Arts and Humanities**]: *Music.*

General Terms
Measurement, Performance, Experimentation.

1. INTRODUCTION
Music digital libraries (MDL) face the challenge of providing users with natural and diversified access points to music. Music mood has been recognized as an important criterion when people organize and access music objects [27]. The ever growing amount of music data in large MDL systems calls for the development of automatic tools in classifying music by mood. To date, most automatic music mood classification algorithms and systems are solely based on the audio content of music (as recorded in .wav, .mp3 or other popular formats) (e.g., [16][25]). In recent years, researchers have started to exploit music lyrics in mood classification (e.g., [9][11]) and hypothesize that lyrics, as a separate source from music audio, might be complementary to audio content. Hence, some researchers have started to combine lyrics and audio, and initial results indeed show improved classification performances [13][30].

The work presented in this paper was premised on the belief that audio and lyrics information would be complementary. However, we wondered a) which text features would be the most useful in the task of music mood classification; b) how best the two information sources could be combined; and, c) how much it would help to combine lyrics and audio on a large experimental dataset. There have been quite a few studies on sentiment analysis in the text domain [18], but most recent experiments on combining lyrics and audio in music classification only used basic text features (e.g., content words and part-of-speech).

In this study, we examine and evaluate a wide range of lyric text features including the basic features used in previous studies, linguistic features derived from sentiment lexicons and psycholinguistic resources, and text stylistic features. We attempt to determine the most useful lyric features by comparing various lyric feature types and their combinations. The best lyric features are then combined with a leading audio-based mood classification system, MARSYAS [26], using two fusion methods that have been used successfully in other music classification experiments: feature concatenation [13] and late fusion of classifiers [4]. After determining the best hybrid system, we then compare the performances of the hybrid systems against lyric-only and audio-only systems. The learning curves of these systems are also compared in order to find out whether adding lyrics can help reduce training data required for effective mood classification.

This study contributes to the MDL research domain in two novel and significant ways:

1) Many of the lyric text features examined here have never been formally studied in the context of music mood classification. Similarly, most of the feature type combinations have never previously been compared to each other using a common dataset. Thus, this study pushes forward the state-of-the-art on sentiment analysis in the music domain;

2) The ground truth dataset built for this study is unique. It contains 5,296 unique songs in 18 mood categories derived from social tags. This is one of the largest experimental datasets in music mood classification with ternary information sources available: audio, lyrics and social tags. Part of the dataset has been made available to the MDL and Music Information Retrieval (MIR) communities through the 2009 iteration of the Music Information Retrieval Evaluation eXchange (MIREX) [7], a

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community-based framework for the formal evaluation of algorithms and techniques related to MDL and MIR development.

The rest of the paper is organized as follows. Related work is critically reviewed in Section 2. Section 3 introduces the various lyric features we examined and evaluated. Our experiment design is described in Section 4, including the dataset, the audio-based system, and the evaluation task and measures. In Section 5 we present the experimental results and discuss issues raised from them. Section 6 concludes the paper and proposes future work.

2. RELATED WORK

2.1 Music Mood Classification Using Single Sources

Most existing work on automatic music mood classification is exclusively based on audio features among which spectural and rhythmic features are the most popular across studies (e.g.,[16][19][25]). The datasets used in these experiments usually consisted of several hundred to 1,000 songs labeled with four to six mood categories.

Very recently, studies on music mood classification solely based on lyrics have appeared [9][11]. In [9], the authors compared traditional bag-of-words features in word unigrams, bigrams, trigrams and their combinations, as well as three feature representation models (i.e., Boolean, absolute term frequency and tfidf weighting). Their results showed that the combination of unigrams, bigrams and trigrams with tfidf weighting performed the best, indicating higher-order bag-of-words features captured more semantics useful for mood classification. The authors of [11] moved beyond bag-of-words lyric features, and extracted features based on an affective lexicon translated from the Affective Norm of English Words (ANEW) [5]. The datasets used in both studies were relatively small: the dataset in [9] contained 1,903 songs in only two mood categories, “love” and “lovelorn”, while [11] classified 500 Chinese songs into four mood categories derived from Russell’s arousal-valence model [20].

From a different angle, [3] tried to use social tags to predict mood and theme labels of popular songs. The authors designed the experiments as a tag recommendation task where the algorithm automatically suggested mood or theme descriptors given social tags associated with a song. Although they used 6,116 songs and 89 mood-related descriptors, their study was not comparable to ours in that it did not consider music audio and it was a recommendation task where only the first N descriptors were evaluated ($N = 3$ in [3]).

2.2 Music Mood Classification Combining Text and Audio

The early work combining lyrics and audio in music mood classification can be traced back to [29] where the authors used both lyric bag-of-words features and the 182 psychological features proposed in the General Inquirer [21] to disambiguate categories that audio-based classifiers found confusing. Although the overall classification accuracy was improved by 2.1%, their dataset was too small (145 songs) to draw any reliable conclusions. Laurier et al. [13] also combined audio and lyric bag-of-words features. Their experiments on 1,000 songs in four categories (also from Russell’s model) showed that the combined features with audio and lyrics improved classification accuracies in all four categories. Yang et al. [30] evaluated both unigram and bigram bag-of-words lyric features as well as three methods for fusing lyric and audio sources on 1,240 songs in four categories (again from Russell’s model) and concluded that leveraging lyrics could improve classification accuracy over audio-only classifiers.

As a very recent work, [4] combined social tags and audio in music mood and theme classification. The experiments on 1,612 songs in four and five mood categories showed that tag-based classifiers performed better than audio-based classifiers while the combined classifiers were the best. Again, it suggested that combining heterogeneous resources helped improve classification performances. Instead of concatenating two feature sets like most previous research did, [4] combined the tag-based classifier and audio-based classifier via linear interpolation (one variation of late fusion), since the two classifiers were built with different classification models. In our study, we use the same classification model for both audio-based and lyric-based classifiers so that we can compare the two fusion methods: feature concatenation and late fusion.

The aforementioned studies used two to six mood categories which were most likely oversimplified and might not reflect the reality of music listening since the categories were adopted from psychology models developed in laboratory settings [10]. This study uses mood categories derived from social tags (see Section 4.1) which have good connections to the reality and are more complete than the commonly used Russell’s model [10]. Furthermore, previous studies used relatively small datasets and only evaluated a few of the most common lyric feature types. It should be noted that the performances of these studies were not comparable because they all used different datasets.

Our previous work [12] went one step further and evaluated bag-of-words lyric features on content words, part-of-speech and function words, as well as three feature selection methods in combining lyric and audio features (F-score feature ranking, SVM feature ranking, and language model comparisons). The experiments were conducted on a substantial subset of the dataset used in this study. The results from this earlier study were quite encouraging. However, we did not 1) examine lyric features based on linguistic resources and text stylistic features; 2) exhaustively compare any combinations of the feature types; 3) evaluate late fusion method in combining lyrics and audio; and, 4) compare the learning curves of the systems.

2.3 Music Genre Classification Combining Text and Audio

Besides mood classification, the combination of audio and lyrics has also been applied to music genre classification [17]. In addition to bag-of-words and part-of-speech features, [17] also proposed novel lyric features such as rhyme features and text stylistic features. In particular, the authors demonstrated interesting distribution patterns of some exemplar lyric features across different genres. For example, the words “nuh” and “fi” mostly occurred in reggae and hip-hop songs. Their experiments on a dataset of 3,010 songs and 10 genre categories showed that the combination of text stylistic features, part-of-speech features and audio spectrum features significantly outperformed the classifier using audio spectrum features only as well as the classifier combining audio and bag-of-words lyric features. This work gives us the insight that text stylistic features may also be useful in mood classification, and thus we include text stylistic features in this study as well.

2.4 Classification Models

Except for [11], which used fuzzy clustering, most of the aforementioned studies used standard supervised learning models such as K-Nearest Neighbors (KNN), Naïve Bayes and Support Vector Machines (SVM). Among them, SVM seem the most popular model with top performances. Therefore, in this study, we use SVM to build classifiers using single or multiple sources.

3. LYRIC FEATURES

In this section, we describe the various lyric feature types we evaluated in this study: 1) basic text features that are commonly used in text categorization tasks; 2) linguistic features based on psycholinguistic resources; and, 3) text stylistic features including those proved useful in [17]. Finally, we describe the 255 combinations of these feature types that were also evaluated in this study.

3.1 Basic Lyric Features

As a starting point, our previous research [12] evaluated bag-of-words features with the following types:

1) content words (Content): all words except function words, without stemming;
2) content words with stemming (Cont-stem): stemming means combining words with the same roots;
3) part-of-speech (POS) tags: such as noun, verb, proper noun, etc. We used the Stanford POS tagger[1] which tagged each word with one of 36 unique POS tags;
4) function words (FW): as opposed to content words, also called "stopwords" in text information retrieval.

For each of the feature types, four representation models were compared: 1) Boolean; 2) term frequency; 3) normalized frequency; and, 4) tfidf weighting.

In this study, we continued to evaluate these bag-of-words features, but also included bigrams and trigrams of these features and representation models. For each *n*-gram feature type, features that occurred less than five times in the training dataset were discarded. Also, for bigrams and trigrams, function words were not eliminated because content words are usually connected via function words as in "I love you" where "I" and "you" are function words.

Theoretically, high order *n*-grams can capture features of phrases and compound words. A previous study on lyric mood classification [9] found the combination of unigrams, bigrams and trigrams yielded the best results among all *n*-gram features. Hence, in this study, we also combine unigrams and bigrams, then unigrams, bigrams and trigrams to see the effect of progressively expanded feature sets. The basic lyric feature sets evaluated in this study are listed in Table 1.

The effect of stemming on *n*-gram dimensionality reflects the unique characteristics of lyrics. The reduction rate of stemming was 3.3% for bigrams and 0.2% for trigrams which were very low compared to other genres of text. An examination of the lyric text suggested that the repetitions frequently used in lyrics indeed made a difference in stemming. For example, the lines, *"bounce bounce bounce"* and *"but just bounce bounce bounce, yeah"* were stemmed to *"bounc bounc bounc"* and *"but just bounc bounc bounce, yeah"*. The original bigram *"bounce bounce"* and trigram *"bounce bounce bounce"* then expanded into two bigrams and two trigrams after stemming.

Table 1. Summary of basic lyric features

Feature Type	*n*-grams	No. of dimensions
Content words without stemming (Content)	unigrams	7,227
	bigrams	34,133
	trigrams	42,795
	uni+bigram	41,360
	uni+bi+trigram	84,155
Content words with stemming (Cont-stem)	unigrams	6,098
	bigrams	33,008
	trigrams	42,707
	uni+bigram	39,106
	uni+bi+trigram	81,813
Part-of-speech (POS)	unigrams	36
	bigrams	1,057
	trigrams	8,474
	uni+bigram	1,093
	uni+bi+trigram	9,567
Function words (FW)	unigrams	467
	bigrams	6,474
	trigrams	8,289
	uni+bigram	6,941
	uni+bi+trigram	15,230

3.2 Linguistic Lyric Features

In the realm of text sentiment analysis, domain dependent lexicons are often consulted in building feature sets. For example, Subasic and Huettner [23] manually constructed a word lexicon with affective scores for each affect category considered in their study, and classified documents by comparing the average scores of terms included in the lexicon. Pang and Lee [18] summarized that studies on text sentiment analysis often used existing off-the-shelf lexicons or automatically built problem-dependent lexicons using supervised methods. While the three feature ranking and selection methods used in our previous work [12] (i.e., F-Score, SVM score and language model comparisons) were examples of supervised lexicon induction, in this study, we focus on features extracted by using a range of psycholinguistic resources: General Inquirer (GI), WordNet, WordNet-Affect and Affective Norm of English Words (ANEW).

3.2.1 Lyric Features based on General Inquirer

General Inquirer (GI) is a psycholinguistic lexicon containing 8,315 unique English words and 182 psychological categories [20]. Each sense of the 8,315 words in the lexicon is manually labeled with one or more of the 182 psychological categories to which the sense belongs. For example, the word "happiness" is associated with the categories "Emotion", "Pleasure", "Positive", "Psychological well being", etc. The mapping between words and psychological categories provided by GI can be very helpful in looking beyond word forms and into word meanings, especially for affect analysis where a person's psychological state is exactly

[1] http://nlp.stanford.edu/software/tagger.shtml

the subject of study. One of the related studies on music mood classification [29] used GI features together with lyric bag-of-words and suggested representative GI features for each of their six mood categories.

GI's 182 psychological features are also evaluated in our current study. It is noteworthy that some words in GI have multiple senses (e.g., "happy" has four senses). However, sense disambiguation in lyrics is an open research problem that can be computationally expensive. Therefore, we merged all the psychological categories associated with any sense of a word, and based the match of lyric terms on words, instead of senses. We represented the GI features as a 182 dimensional vector with the value at each dimension corresponding to either word frequency, tfidf, normalized frequency or Boolean value. We denote this feature type as "GI".

The 8,315 words in General Inquirer comprise a lexicon oriented to the psychological domain, since they must be related to at least one of the 182 psychological categories. We then built bag-of-words features using these words (denoted as "GI-lex"). Again, we considered all four representation models for this feature type which has 8,315 dimensions.

3.2.2 Lyric Features based on ANEW and WordNet

Affective Norms for English Words (ANEW) is another specialized English lexicon [5]. It contains 1,034 unique English words with scores in three dimensions: *valence* (a scale from unpleasant to pleasant), *arousal* (a scale from calm to excited), and *dominance* (a scale from submissive to dominant). All dimensions are scored on a scale of 1 to 9. The scores were calculated from the responses of a number of human subjects in psycholinguistic experiments and thus are deemed to represent the general impression of these words in the three affect-related dimensions. ANEW has been used in text affect analysis for such genres as children's tales [1] and blogs [15], but the results were mixed with regard to its usefulness. In this study, we strive to find out whether and how the ANEW scores can help classify text sentiment in the lyrics domain.

Besides scores in the three dimensions, for each word ANEW also provides the standard deviation of the scores in each dimension given by the human subjects. Therefore there are six values associated with each word in ANEW. For the lyrics of each song, we calculated means and standard deviations for each of these values of words included in ANEW, which gave us 12 features.

As the number of words in the original ANEW is too few to cover all the songs in our dataset, we expanded the ANEW word list using WordNet [8]. WordNet is an English lexicon with marked linguistic relationships among word senses. It is organized by *synsets* such that word senses in one *synset* are essentially synonyms from the linguistic point of view. Hence, we expanded ANEW by including all words in WordNet that share the same *synset* with a word in ANEW and giving these words the same ANEW scores as the one in ANEW. Again, we did not differentiate word senses since ANEW only presents word forms without specifying which sense is used. After expansion, there are 6,732 words in the expanded ANEW which covers all songs in our dataset. That is, every song has non-zero values in the 12 dimensions. We denote this feature type as "ANEW".

Like the words from General Inquirer, the 6,732 words in the expanded ANEW can be seen as a lexicon of affect-related words. There is another linguistic lexicon of affect-related words called WordNet-Affect [22]. It is an extension of WordNet where affective labels are assigned to concepts representing emotions, moods, or emotional responses. There are 1,586 unique words in the latest version of WordNet-Affect. These words together with words in the expanded ANEW form an affect lexicon of 7,756 unique words. We used this set of words to build bag-of-words features under our four representation models. This feature type is denoted as "Affect-lex".

3.3 Text Stylistic Features

Text stylistic features often refer to interjection words (e.g., "ooh", "ah"), special punctuations (e.g., "!", "?") and text statistics (e.g., number of unique words, length of words, etc.). They have been used effectively in text stylometric analyses dealing with authorship attribution, text genre identification, author gender classification and authority classification [2]. In the music domain, as mentioned in Section 2, text stylistic features on lyrics were successfully used in music genre classification. In this study, we evaluated the text stylistic features defined in Table 2. We initially included all punctuation marks and all common interjection words, but as text stylistic features appeared to be the most interesting feature type in our experiments (see Section 5.2), we also performed feature selection on punctuation marks and interjection words. It turned out that using the top-ranked words and marks (shown in Table 2) yielded the best results. Therefore, throughout this paper, we denote the features listed in Table 2 as "TextStyle" which we compare to other feature types.

Table 2. Text stylistic features evaluated in this study

Feature	Definition
interjection words	normalized frequencies of "hey", "ooh", "yo", "uh", "ah", "yeah"
special punctuation marks	normalized frequencies of "!", "-"
NUMBER	normalized frequency of all non-year numbers
No. of words	total number of words
No. of uniqWords	total number of unique words
repeatWordRatio	(No. of words – No. of uniqWords) / No. of words
avgWordLength	average number of characters per word
No. of lines	total number of lines
No. of uniqLines	total number of unique lines
No. of blankLines	number of blank lines
blankLineRatio	No. of blankLines / No. of lines
avgLineLength	No. of words / No. of lines
stdLineLength	standard deviation of number of words per line
uniqWordsPerLine	No. of uniqWords / No. of lines
repeatLineRatio	(No. of lines – No. of uniqLines) / No. of lines
avgRepeatWord RatioPerLine	average repeat word ratio per line
stdRepeatWordRatio PerLine	standard deviation of repeat word ratio per line
No. of words per min	No. of words / song length in minutes
No. of lines per min	No. of lines / song length in minutes

So far we have described the basic text features, linguistic features and text stylistic features. The latter two kinds of features types are summarized in Table 3, with their numbers of dimensions and numbers of representation models.

Table 3. Summary of linguistic and stylistic lyric features

Feature Abbre.	Feature Type	No. of dimensions	No. of representations
GI	GI psychological features	182	4
GI-lex	words in GI	8,315	4
ANEW	scores in expanded ANEW	12	1
Affect-lex	words in expanded ANEW and WordNet-Affect	7,756	4
TextStyle	Text stylistic features	25	1

3.4 Feature Type Concatenations

Combinations of different feature types may yield performance improvements. For example, [17] found the combination of text stylistic features and part-of-speech features achieved better classification performance than using either feature type alone. In this study, we first determine the best representation of each feature type and then the best representations are concatenated with one another. Specifically, for the basic lyric feature types listed in Table 1, the best performing *n*-grams and representation of each type (i.e., content words, part-of-speech, and function words) was chosen and then further concatenated with linguistic and stylistic features.

For each of the linguistic feature types with four representation models, the best representation was selected and then further concatenated with other feature types. In total, there were eight selected feature types: 1) *n*-grams of content word (either with or without stemming); 2) *n*-grams of part-of-speech; 3) *n*-grams of function words; 4) GI; 5) GI-lex; 6) ANEW; 7) Affect-lex; and, 8) TextStyle. The total number of feature type concatenations is $\sum_{i=1}^{8} C_8^i = 255$, where C denotes the combinations of choosing i types from all eight types (i = 1,…,8). All the 255 feature type concatenations as well as individual feature types were compared in our experiments to find out which lyric feature type or concatenation of multiple types would be the best for the task of music mood classification.

4. EXPERIMENTS

A series of experiments were conducted to find out the best lyric features, the best fusion method, and the effect of lyrics in reducing the number of required training samples. This section describes the experimental setup including the dataset, the audio-based system and the evaluation measures.

4.1 Dataset

Our dataset and mood categories were built from an in-house set of audio tracks and the social tags associated with those tracks, using linguistic resources and human expertise. The process of deriving mood categories and building ground truth dataset was described in [12]. In this section we summarize the characteristics of the dataset.

There are 18 mood categories represented in this dataset, and each of the categories comprises 1 to 25 mood-related social tags downloaded from last.fm, one of the most popular social tagging websites for Western music. A mood category consists of tags that are synonyms identified by WordNet-Affect and verified by two human experts who are both native English speakers and respected MIR/MDL researchers. The song pool was limited to those audio tracks at the intersection of being available to the authors, having English lyrics available on the Internet, and having social tags available on last.fm. For each of these songs, if it was tagged with any of the tags associated with a mood category, it was counted as a positive example of that category. In this way, one single song could belong to multiple mood categories. This is in fact more realistic than a single-label setting since a music piece may carry multiple moods such as "happy and calm" or "aggressive and depressed". For example, the song, *I'll Be Back* by the Beatles was a positive example of the categories "calm" and "sad"; while the song, *Down With the Sickness* by Disturbed was a positive example of the categories "angry", "aggressive" and "anxious".

In this study, we adopted a binary classification approach for each of the mood categories. Negative examples of a category were songs that were not tagged with any of the tags associated with this category but were heavily tagged with many other tags. For instance, the rather upbeat song, *Dizzy Miss Lizzy* by the Beatles was selected as a negative example of the categories "gloomy" and "anxious".

Table 4 lists the mood categories[2] and the number of positive songs in each category. We balanced equally the positive and negative set sizes for each category, and the dataset contains 5,296 unique songs in total. This number is much smaller than the total number of samples in all categories (which is 12,980) because categories often share samples.

Table 4. Mood categories and number of positive songs

Category	No. of songs	Category	No. of songs	Category	No. of songs
calm	1,680	angry	254	anxious	80
sad	1,178	mournful	183	confident	61
glad	749	dreamy	146	hopeful	45
romanti	619	cheerful	142	earnest	40
gleeful	543	brooding	116	cynical	38
gloomy	471	aggressive	115	exciting	30

Table 5. Genre distribution of songs in our dataset

Genre	No. of songs	Genre	No. of songs	Genre	No. of songs
rock	3,977	reggae	55	oldies	15
hip hop	214	jazz	40	other	35
country	136	blues	40	unknown	564
electronic	94	metal	37	TOTAL	5,296
r&b	64	new age	25		

The decomposition of genres in this dataset is shown in Table 5. Although the dataset is dominated by Rock music, a closer

[2] Due to space limitations, each category is denoted by one representative tag in Table 4, but a category can contain 1 to 25 tags. The complete tag list can be found at http://www.music-ir.org/archive/figs/18moodcats.htm

examination at the distribution of songs in different genres and moods showed patterns complying with common knowledge on music. For example, all Metal songs are in negative moods, particularly "aggressive" and "angry". Most New Age songs are associated with the moods of "calm", "sad" and "dreamy" while none of them are with "angry", "aggressive" or "anxious".

4.2 Audio-based Features and Classifiers

Previous studies have generally reported lyrics alone were not as effective as audio in music classification [14][17]. To find out whether this is true with our lyric features that have not been previously evaluated, we compared two best performing lyric feature sets (see Section 5.3) to a leading audio-based classification system evaluated in the Audio Mood Classification (AMC) task of MIREX 2007 and 2008: MARSYAS [26]. Because MARSYAS was the top-ranked system in AMC, its performance sets a challenging baseline against which comparisons must be made.

MARSYAS used 63 spectral features: means and variances of Spectral Centroid, Rolloff, Flux, Mel-Frequency Cepstral Coefficients (MFCC), etc. These features are musical surface features based on the signal spectrum obtained by a Short Time Fourier Transformation (STFT). Spectral Centroid is the mean of the spectrum amplitudes, indicating the "brightness" of a musical signal. Spectral Rolloff is the frequency where 85% of the energy in the spectrum resides below. It is an indicator of the skewness of the frequencies in a musical signal. Spectral Flux is spectral correlation between adjacent time windows, and is often used as an indication of the degree of change of the spectrum between windows. MFCC are features widely used in speech recognition, and have been proved effective in approximating the response of the human auditory system.

The MARSYAS system used Support Vector Machines (SVM) as its classification model. Specifically, it integrated the LIBSVM [6] implementation with a linear kernel to build the classifiers.

In our experiments, all the audio tracks in the dataset were converted into 44.1kHz stereo .wav files before audio features were extracted using MARSYAS.

4.3 Evaluation Measures and Classifiers

For each of the experiments, we report the accuracy across categories averaged in a macro manner, giving equal importance to all categories. For each category, accuracy was averaged over a 10-fold cross validation. To determine if performances differed significantly, we chose the non-parametric Friedman's ANOVA test because the accuracy data are rarely normally distributed [7]. The samples used in the tests are accuracies on individual mood categories.

We chose SVM as the classifiers due to their strong performances in both text categorization and music classification tasks. Like MARSYAS, we also used the LIBSVM implementation of SVM. We chose a linear kernel since trial runs with polynomial kernels yielded similar results and were computationally much more expensive. The default parameters were used for all the experiments because they performed best for most cases where parameters were tuned using the grid search tool in LIBSVM.

5. RESULTS

5.1 Best Individual Lyric Feature Types

For the basic lyric features, the variations of uni+bi+trigrams in the Boolean representation worked best for all three feature types (content words, part-of-speech, and function words). Stemming did not make a significant difference on the performances of content word features, but features without stemming had higher averaged accuracy. The best performance of each individual feature type is presented in Table 6.

Table 6. Individual lyric feature type performances

Feature Abbre.	Feature Type	Represe ntation	Accu racy
Content	uni+bi+trigrams of content words	Boolean	**0.617**
Cont-stem	uni+bi+trigrams of stemmed content words	tfidf	0.613
GI-lex	words in GI	Boolean	0.596
FW	uni+bi+trigrams of function words	Boolean	0.594
Affect-lex	words in expanded ANEW and WordNet-Affect	tfidf	0.594
GI	GI psychological features	tfidf	0.586
POS	uni+bi+trigrams of part-of-speech	Boolean	0.579
ANEW	scores in expanded ANEW	-	0.545
TextStyle	text stylistic features	-	0.529

For individual feature types, the best performing one was Content, the bag-of-words features of content words with multiple orders of *n*-grams. Individual linguistic feature types did not perform as well as Content, and among them, bag-of-words features (i.e., GI-lex and Affect-lex) were the best. The poorest performing feature types were ANEW and TextStyle, both of which were statistically different from the other feature types (at $p < 0.05$). There was no significant difference among the remaining feature types.

5.2 Best Combined Lyric Feature Types

The best individual feature types (shown in Table 6 excluding Cont-stem) were concatenated with one another, resulting in 255 combined feature types. Because value ranges of the feature types varied a great deal (e.g., some are counts, others are normalized weights, etc.), all feature values were normalized to the interval of [0, 1] prior to concatenation. Table 7 shows the best combined feature sets among which there was no significant difference (at $p < 0.05$).

Table 7. Best performing concatenated lyric feature types

Type	No. of dimensions	Accuracy
Content+FW+GI+ANEW+Affect-lex +TextStyle	115,091	0.638
ANEW+TextStyle	37	0.637
Content+FW+GI+GI-lex+ANEW +Affect-lex+TextStyle	123,406	0.637
Content+FW+GI+GI-lex+TextStyle	115,638	0.636
Content+FW+GI+Affect-lex+TextStyle	115,079	0.636
Content+FW+GI+TextStyle	107,323	0.635

The best performing feature combination was Content + FW + GI + ANEW + Affect-lex + TextStyle which achieved an accuracy 2.1% higher than the best individual feature type, Content (0.638 vs. 0.617). All of the lyric feature type concatenations listed in Table 7 contained text stylistic features (TextStyle), although TextStyle performed the worst among all individual feature types (as shown in Table 6). This indicates that TextStyle must have captured very different characteristics of the data than other feature types and thus could be complementary to others. The top three feature combinations also contain ANEW scores, and ANEW alone was also significantly worse than other individual feature types (at $p < 0.05$). It is interesting to see that two poorest performing feature types scored second best when combined with each other. In addition, the ANEW and TextStyle feature types are the only two types that do not conform to the bag-of-words framework among all eight individual feature types.

Except for the combination of ANEW and TextStyle, all of the other top performing feature combinations shown in Table 7 are concatenations of four or more feature types, and thus have very high dimensionality. In contrast, ANEW+TextStyle has only 37 dimensions, which is certainly a lot more efficient than the others. On the other hand, high dimensionality provides room for feature selection and reduction. Indeed, our previous work in [12] applied three feature selection methods on basic unigram lyric features and showed improved performances. We leave it to future work to investigate feature selection and reduction for combined feature sets with high dimensionality.

Except for ANEW+TextStyle, all other top performing feature concatenations contained the combination of Content, FW, GI and TextStyle. In order to see the relative importance of the four individual feature types, we compared the combinations of any three of the four types in Table 8.

Table 8. Performance comparison of Content, FW, GI and TextStyle

Type	Accuracy
Content+FW+TextStyle	0.632
Content+FW+GI	0.631
Content+GI+TextStyle	0.624
FW+GI+TextStyle	0.619

The combination of FW+GI+TextStyle performed the worst among the combinations shown in Table 8. Together with the fact that Content performed the best among all individual feature types, we can safely state that content words are still very important in the task of lyric mood classification.

5.2.1 *Analysis of Text Stylistic features*

As TextStyle is a very interesting feature type, we took a closer look at it to determine the most important features within this type. As mentioned in Section 3.3, we initially included all punctuation marks and common interjection words in this feature type, and then we ranked and selected the *n* most important interjection words and punctuation marks (denoted as “I&P” in Table 9). We kept the 17 text statistic features defined in Table 2 (denoted as “TextStats” in Table 9) unchanged in this set of experiments because the 17 dimensions of text statistics were already compact compared to the 134 interjection words and punctuations. Since we used SVM as the classifier, and a previous study [31] suggested feature selection using SVM ranking worked best for SVM classifiers, we ranked the features according to the feature weights calculated by the SVM classifier and compared the performances using varied numbers of top-ranked features. Like all experiments in this paper, the results were averaged across a 10-fold cross validation, and the feature selection was performed only using training data in each fold. Table 9 shows the results, from which we can see that many of the interjection words and punctuation marks are redundant indeed.

Table 9. Feature selection for TextStyle

No. of features			Accuracy	
TextStats	I&P	Total	TextStyle	ANEW+TextStyle
17	0	17	0.524	0.634
17	8	25	0.529	**0.637**
17	15	32	0.526	0.632
17	25	42	0.514	0.631
17	45	62	0.514	0.628
17	75	92	0.513	0.615
17	134	151	0.513	0.612

To provide a sense of how the top features distributed across the positive and negative samples of the categories, we plotted distributions for each of the selected TextStyle features. Figures 1-3 illustrate the distributions of three sample features: “hey”, “!”, and “number of words per minutes”. As can be seen from the figures, the positive and negative bars for each category generally have uneven heights. The more different they are, the more distinguishing power the feature would have for that category.

5.3 Best Fusion Method

Since the best lyric feature set was Content + FW + GI + ANEW + Affect-lex + TextStyle (denoted as “BEST” thereafter), and the second best feature set, ANEW + TextStyle was very interesting, we combined each of the two lyric feature sets with the audio-based system described in Section 4.2. Fusion methods can be used to flexibly integrate heterogeneous data sources to improve classification performance, and they work best when the sources are sufficiently diverse and thus can possibly make up for each other's mistakes. Previous work in music classification has used such hybrid sources as audio and social tags, audio and lyrics, etc. There are two popular fusion methods. The most straightforward one is feature concatenation where the two feature sets are concatenated and the classification algorithms run on the combined feature vectors (e.g., [13][17]). The other method is often called “late fusion” which is to combine the outputs of individual classifiers based on different sources, either by (weighted) averaging (e.g., [28][4]) or by multiplying (e.g., [14]).

According to [24], in the case of combining two classifiers for binary classification as in this research, the two late fusion variations, averaging and multiplying are essentially the same. Therefore, in this study we used the weighted averaging-estimation. For each testing instance, the final estimation probability was calculated as:

$$p_{hybrid} = \alpha\, p_{lyrics} + (1-\alpha)\, p_{audio} \tag{2}$$

where α is the weight given to the posterior probability estimated by the lyric-based classifier. A song was classified as positive when the hybrid posterior probability was larger or equal than 0.5. We varied α from 0.1 to 0.9 with an increment step of 0.1, and the average accuracies with different α values are shown in Figure 4.

As Figure 4 shows, the highest average accuracy was achieved when $\alpha = 0.5$ for both lyric feature sets, that is when the lyric-based and audio-based classifiers got equal weights.

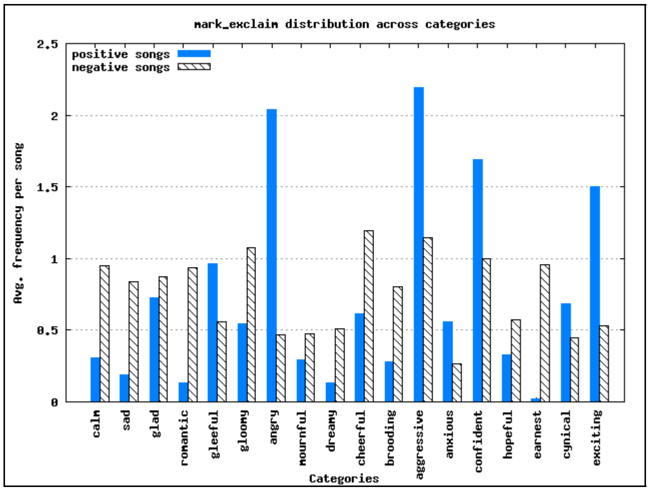

Figure 1. Distributions of "!" across categories

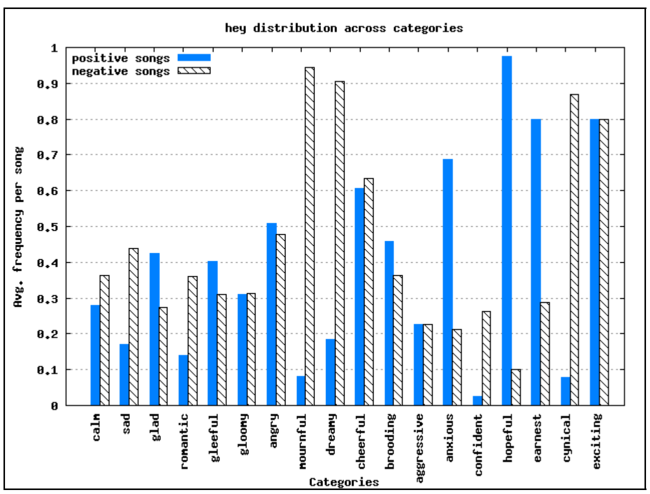

Figure 2. Distributions of "hey" across categories

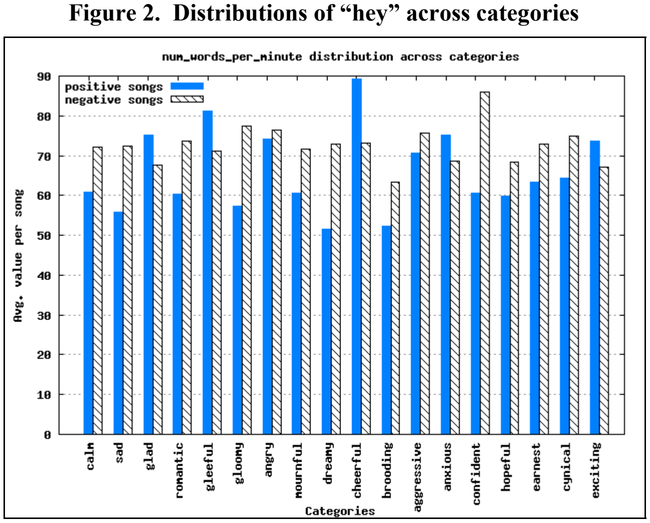

Figure 3. Distributions of average values of "number of words per minutes" across categories

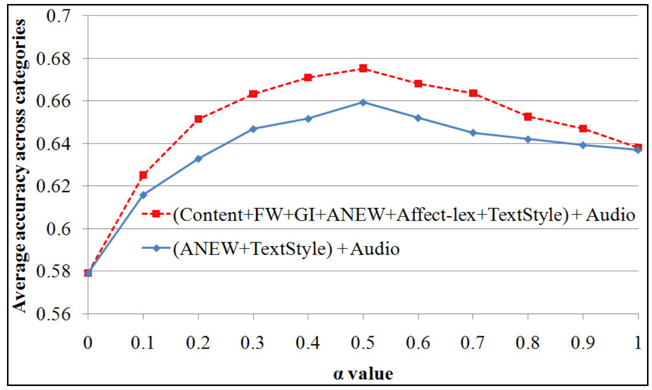

Figure 4. Effect of α value in late fusion on averaged accuracy

Table 10 presents the average accuracies of single-source-based systems and hybrid systems with the aforementioned two fusion methods. It is clear from Table 10 that feature concatenation was not good for combining ANEW+TextStyle feature set and audio. Late fusion was a good method for both lyric feature sets but again, the BEST lyric feature combination outperformed ANEW + TextStyle in combining with audio (0.675 vs. 0.659) with a statistically insignificant difference ($p < 0.05$). Table 11 shows the results of pair-wise statistical tests on system performances for both lyric feature sets.

Table 10. Accuracies of single-source and hybrid systems

Feature set	Audio-only	Lyric-only	Feature concatenation	Late fusion
BEST	0.579	0.638	0.645	0.675
ANEW+Style	0.579	0.637	0.629	0.659

Table 11. Statistical tests on pair-wise system performances for BEST and ANEW+TextStyle lyric feature set

Feature set	Better System	Worse System	*p* value
BEST	Late fusion	Audio-only	**0.001**
BEST	Feature concate.	Audio-only	**0.027**
BEST	Lyrics-only	Audio-only	0.054
BEST	Late fusion	Lyrics-only	0.110
BEST	Feature concate.	Lyrics-only	0.517
BEST	Late fusion	Feature concate.	0.327
ANEW+Style	Late fusion	Audio-only	**0.004**
ANEW+Style	Feature concate.	Audio-only	**0.045**
ANEW+Style	Lyrics-only	Audio-only	0.074
ANEW+Style	Late fusion	Lyrics-only	0.217
ANEW+Style	Late fusion	Feature concate.	0.569
ANEW+Style	Lyrics-only	Feature concate.	0.681

The statistical tests showed that both hybrid systems using late fusion and feature concatenation were significantly better than the audio-only system at $p < 0.05$. In particular, the hybrid systems with late fusion improved accuracy over the audio-only system by 9.6% and 8% for the top two lyric feature sets respectively. These showed the usefulness of lyrics in complementing music audio in the task of mood classification. Within the two hybrid systems, late fusion outperformed feature concatenation for 3%, but the differences were not statistically significant. Besides, the raw differences around 5.9% between the performances of the lyric-only systems and the audio-only system are noteworthy. The findings of other researchers have never shown lyric-only systems to outperform audio-only systems in terms of averaged accuracy across all categories[13][17][30]. We surmise that this difference could be because of the new lyric features applied in this study.

Figure 5 shows the system accuracies across individual mood categories for the BEST lyric feature sets where the categories are in descending order of the number of songs in each category.

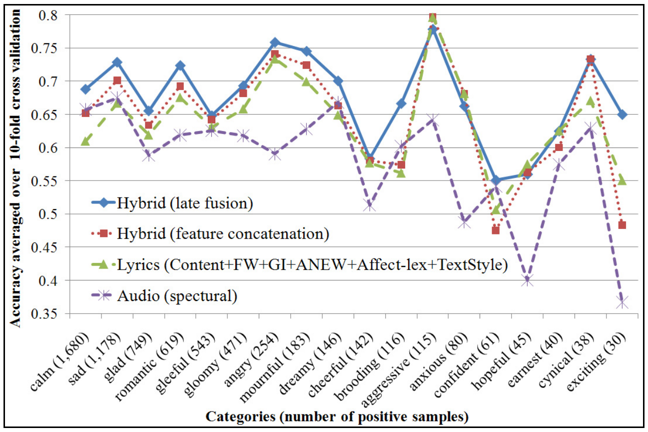

Figure 5. System accuracies across individual categories

Figure 5 reveals that system performances become more erratic and unstable after the category "cheerful". Those categories to the right of "cheerful" have few than 142 positive examples. This suggests that the systems are vulnerable to the data scarcity problem. Also worthy of future investigation is the examination of those categories where the audio-only system did outperform the lyric-only system: "calm", "brooding", and "confident".

Given the high performance of Content lyric features, Table 12 lists the top five Content features in selected categories. For categories where lyric features outperformed audio features, the top *n*-grams seem to have intuitively meaningful connections to the categories, such as "with you" in "romantic" songs and "happy" in "cheerful" songs. However, there is no such semantic connection for "calm" where audio outperformed lyric features.

Table 12. Top five Content features in selected categories

romantic	cheerful	hopeful	angry	aggressive	calm
with you	i love	you ll	baby	fuck	you all
on me	night	strong	i am	dead	all look
crazy	ve got	i get	shit	i am	all look at
come on	happy	loving	scream	girl	you all i
i said	for you	dreams	to you	man	burning

5.4 Learning Curves

In order to find out whether lyrics can help reduce the amount of training data required for achieving certain performance levels, we examined the learning curves of the single-source-based systems and the late fusion hybrid system for the BEST lyric feature set. Presented in Figure 6 are the accuracies of the systems when the number of training samples varied from 10% to 100% of all available training samples.

Figure 6 shows a general trend that all system performances increased with more training data, but the performance of the audio-based system increased much more slowly than the other systems. With 20% training samples, the accuracies of the hybrid and the lyric-only systems were already better than that of the audio-only system with all available training data. To achieve similar accuracy, the hybrid system needed about 20% fewer training examples than the lyric-only system. This validates the hypothesis that combining lyrics and audio can reduce required training samples needed to achieve certain classification performance levels. In addition, the learning curve of the audio-only system levels off at 80% training sample size, while the curves of the other two systems never level off. This indicates the hybrid system and lyric-only system may further improve their performances if given more training examples.

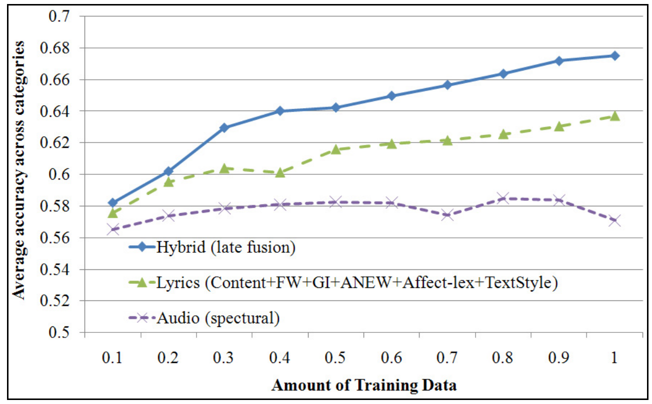

Figure 6. Learning curves of hybrid and single-source systems

6. CONCLUSIONS AND FUTURE WORK

This study evaluated a number of lyric text features in the task of music mood classification, including the basic, commonly used bag-of-words features, features based on psycholinguistic resources and text stylistic features. The experiments on a large dataset revealed that the most useful lyric features were a combination of content words, function words, General Inquirer psychological features, ANEW scores, affect-related words and text stylistic features. A surprising finding was that the combination of ANEW scores and text stylistic features, with only 37 dimensions, achieved the second best performance among all feature types and combinations (compared to 115,091 in the top performing lyric feature combination). In combining lyrics and music audio, late fusion (linear interpolation with equal weights to both classifiers) yielded the best performance, and outperformed a leading audio-only system on this task by 9.6%. Experiments on learning curves discovered that complementing audio with lyrics could reduce the number of training samples required to achieve the same or better performance than single-source-based systems. These findings can help improve the effectiveness and efficiency of music mood classification and thus pave the way to making mood a practical and affordable access point in Music Digital Libraries.

As a direction of future work, the interaction of features and classifiers is worthy of further investigation. Using classification models other than SVM (e.g., Naïve Bayes), the top-ranked features might be different than those selected by SVM. With proper feature selection methods, other classification models might outperform SVM.

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